LOFAR PASSIVE-SONAR PROCESSING FOR MARINE VESSEL DETECTION

From underwater acoustic noise to time–frequency display (LOFARgram)

1. Underwater sources (radiated noise)



Vessels radiate broadband and tonal noise from machinery, propellers and other sources, which propagates through the marine environment to a hydrophone or towed array.

2. Hydrophone / array and analogue conditioning



Received signal is amplified and band-limited to prevent aliasing before digitisation.

3. Sampling (ADC)

Analogue to digital conversion

ADC

Sample at rate f_s (≥ 2 × highest frequency of interest).

4. Windowed time frames

Split into short, overlapping frames (e.g. 1–10 s)



Apply a window function (e.g. Hanning) to each frame to reduce spectral leakage.

5. FFT

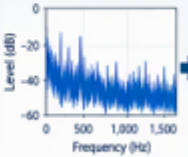
Compute Fast Fourier Transform for each frame

FFT

Convert time-domain data to frequency domain for each frame.

6. Magnitude / power spectrum

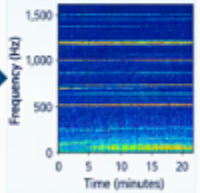
Spectrum for each frame



Display magnitude (or power) spectrum, usually in dB.

7. LOFARgram (time–frequency display)

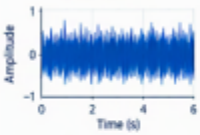
Stack spectra over time



Sequential spectra are stacked to show how frequency content changes over time (a LOFARgram).

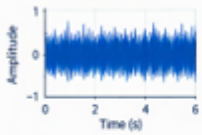
Example signals at each stage

Received time-domain waveform (at hydrophone)



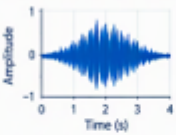
Mixture of broadband noise and tonal components.

After analogue conditioning (band-limited signal)



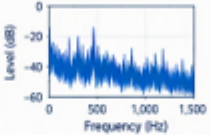
Filtered signal (e.g. 10 Hz – 1.5 kHz).

Windowed frame
(e.g. 4 s with Hanning window)



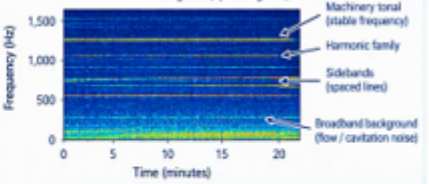
Window reduces spectral leakage.

Magnitude spectrum
(for one frame)



Tonal lines above broadband background.

LOFARgram (spectrogram)



LOFARgram orientation

- **Vertical axis:** frequency (Hz) — low frequencies (LOFAR) are at the bottom.
- **Horizontal axis:** time (seconds or minutes).
- **Colour:** signal level (dB) — warmer colours = higher level.
- **Result:** persistent horizontal lines indicate stable tonals; vertical changes show frequency shifts with time (e.g. manoeuvring, speed change).

Typical features in a LOFARgram

- **Machinery tonals:** stable horizontal lines (e.g. engine orders).
- **Harmonic families:** integer multiples of a fundamental.
- **Propeller/shaft related tonals:** rotation rate and harmonics (often low frequency).
- **Sidebands:** lines either side of a tonal, spaced by modulation frequencies (e.g. shaft rate or blade-pass rate).
- **Broadband background:** flow and cavitation noise, increasing with speed.

Key takeaways

- LOFAR processing reveals narrowband tonal structure in the acoustic spectrum over time.
- Persistent tonals and harmonic patterns enable detection, classification and monitoring of surface ships and submarines.
- LOFAR is complementary to DEMON (modulation spectrum) and cepstrum (period detection) analysis.